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Graduation Work/Thesis Final Report

2025 year 1 semester

Title : Textual Analysis for Automated Age Rating of
Video Games From Transcript Files

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■ Abstract (Write the abstract of the report)

This graduation work presents a novel two-stage pipeline approach for automated prediction of ESRB ratings from video game transcripts. Unlike existing methods that rely on manual content evaluation or predefined descriptors, our system combines GPT-4o-mini's natural language understanding with a Random Forest classifier to analyze raw game text and predict appropriate age ratings. The system first uses GPT to identify the presence of 31 ESRB content descriptors in game transcripts, then processes this binary feature vector through a Random Forest model trained on over 2,000 games with known ratings. Evaluation across 40 diverse game transcripts spanning from 1990-2024 demonstrated consistently high accuracy (81.7%), with perfect prediction for Mature-rated games (100%) and strong performance for Everyone-rated titles (93.3%). The implemented web application provides comprehensive analysis including rating predictions with confidence scores, content severity percentages across different dimensions, and evidence extraction with contextual examples from the transcript. Additional features include a dedicated summarizer for game content exploration and an alternative questionnaire-based prediction approach. This work addresses critical industry challenges of resource intensity, cost, and consistency in content rating, demonstrating significant potential for practical application in the gaming ecosystem.

■ Introduction (no more than 6 page)

A. Background of proposal and need

The video game industry, once small compared to other entertainment industries, experienced rapid growth in recent years, reaching a global market value of \$184.3 billion in 2024 [1] and it continues to grow to this day. This is mostly due to the increasing number of gamers in the world, and the number of video game users is estimated to reach 3.02 billion users in 2029 [2]. It means that more people, especially young people, are getting exposed to video games with a wide range of content, from harmless games to games filled with intense violence. On Steam alone, the largest digital distribution platform for PC games, 18,890 games were released in 2024 [3]. And out of all games released in 2024, 12% of games were rated as Mature [4], making them unsuitable for individuals under the age of 18. Considering that around 20% of all gamers are under 18 [5], there is a big risk that children might be exposed to unsuitable content for them. That strengthens the importance of a proper game content

rating system that effectively protects children from inappropriate content while educating parents about the nature of games their children might access.

Video games have gone a long way and with evolution they had, the content shown in the games has also matured with hundreds of games releasing every year that are specifically intended for adult audiences. In 1994, the Entertainment Software Rating Board (ESRB) was founded by the Interactive Digital Software Association [6] as a self-regulatory organization to provide age and content ratings for video games in the United States and Canada. ESRB ratings provide information about what's in a game or app so parents and consumers can make informed choices about which games are right for their family [7]. The ESRB rating system consists of 5 categories: E (Everyone), E10+ (Everyone 10 and older), T (Teen), M (Mature 17+), and AO (Adults Only 18+), with the first four being the main ones and the most commonly assigned.

The current approach for assigning age ratings is manual, with human reviewers manually going through game content. This approach has several disadvantages. One of the main ones is that the current process is time-consuming and labor-intensive. More and more games are getting released every year, and they become bigger and longer, making it harder to assign ratings to all games that are coming out. The strain is not only on the ESRB, but on the publishers, since they have to provide a completed questionnaire detailing any relevant content (violence, sex, language, gambling, etc.), and other factors such as context, reward systems, and player control, and a video showing typical gameplay, missions, and cutscenes, including the most "extreme" content [8]. Such resource intensity makes the process not cheap. To get a retail rating, publishers should pay \$3,000, and since big companies like Sony require an ESRB rating, it puts a big financial strain on small game developers [9]. There are also big consistency concerns since the current approach is prone to human bias, which can introduce potential inconsistencies in how content is evaluated across different reviewers and over time.

Recent advances in artificial intelligence (AI), particularly in natural language processing (NLP) and machine learning, present an opportunity to address the challenges mentioned above. AI is already widely used in social media to automatically identify violations of specific policy lines and remove potentially harmful content like deepfakes or photos with nudity [10]. Moreover, NLP has made significant progress in sentiment analysis, with state-of-the-art models achieving high accuracy (often above 90%) on specific, well-defined datasets and tasks [11]. The machine learning models have also become

increasingly sophisticated at classifying content with awareness of context by being able to understand relationships between words [12]. However, despite these advances, AI technologies are still not being commonly applied to game content rating.

Considering the challenges and the current technological advances, there is an opportunity to develop an automated game content rating prediction system. Such a system would provide many benefits. The most important one is that the rating process will become less human resource-intensive, making it more affordable and scalable. With an automated system more games will be able to get a rating in a faster manner. Also, this approach can increase consistency by providing standardized evaluations using consistent criteria across all games. Such a system can also be made transparent by offering clear explanations for each decision made with specific and direct evidence from the game itself. The system can also be more adaptable since it is easier to change a program than to educate the review board about always changing content standards, cultural contexts, and rules. However, an effective automated rating system should not simply count swear words or violent scenes. It must understand the context in which sensitive content appears and how it is being presented in the narrative of the game.

B. Goal of the graduation work

The primary goal of this work is to create an automated system for predicting ESRB ratings for video games based on their transcripts. The system should be both accurate and transparent to ensure its trustworthiness. This goal addresses the challenges in the background section of the introduction, since the automated system will be able to solve issues like resource efficiency, cost, scalability, and consistency. Previous approaches to predicting age rating for video games include a lexicon-based approach with semantic expansion [13] and machine learning on pre-existing content descriptors [14]. Unlike both previous approaches, which rely on either manually crafted dictionaries or pre-labeled content descriptors, it was determined that this complex problem requires a novel approach that should rely on raw game transcript analysis.

The system should be able to analyze video game transcripts and predict appropriate age ratings with high accuracy. Achieving an accuracy of 80% or more is one of the main priorities of this project to prove the system's effectiveness. The system should also be accurate at distinguishing between the four main ESRB rating categories, which are E, E10+, T, and M. It should

function independently and not need pre-assigned content descriptors or extensive manual intervention. The system would be an end-to-end solution for its users that would require only a transcript of the game and nothing more. Moreover, unlike many AI systems, which are essentially black boxes, the system should be transparent, meaning it should show evidence of why a specific rating was assigned. This includes identifying specific content descriptors present in the game, providing example text from the transcript that demonstrates these descriptors, and offering confidence metrics for the predictions.

The developed approach should be able to understand the context, severity, and presentation of potentially sensitive content across multiple dimensions such as violence, language, substances, and suggestive content. It should replicate how human reviewers analyze video games. This also includes the ability to recognize between mild and strong representations of content types in the transcript. Moreover, not only the presence of the concerning content should be detected, but also the intensity of such content across the main dimensions mentioned earlier should be estimated. This enables a user to understand exactly why a certain rating was assigned.

The created system should be able to demonstrate its practical utility for the gaming industry. The system should be valuable to game developers by providing an initial rating assessment and prediction for their game before the official submission to ESRB. The system should also be potentially helpful not only to game developers, content moderators, and parents, but also to rating organizations. Since instead of analyzing the whole game by themselves, ESRB reviewers will be able to just check the automated system's prediction and its evidence, and then come up with a conclusion from there. It can potentially save resources, time, and effort of the review board. The system should have the potential to make age-appropriate content evaluation more accessible to developers, increase consistency of decisions, provide more detailed, evidence-backed information about game content, and demonstrate a novel approach for video games age rating prediction.

C. Overview of the whole graduation work

This graduation work involves developing an automated system for predicting ESRB ratings based on video game transcripts. Following an extensive literature review on content rating systems and natural language processing techniques, several approaches were tested to address this complex challenge.

Initially, a BERT-based deep learning approach was attempted, expecting it to recognize language patterns associated with different age ratings like

violent descriptions, mature themes, and kid-friendly content. Despite optimizing hyperparameters, this method achieved only 40% accuracy on test data due to overfitting from limited training samples. The dataset created from raw transcripts divided into labeled chunks was not well-suited for training sophisticated models like BERT, resulting in the model memorizing text patterns rather than learning meaningful content relationships. Similarly, applying Random Forest directly on text chunks achieved only 50% accuracy, as the extracted features were not sufficiently informative to distinguish between rating categories.

After thorough experimentation and analysis, a novel two-stage pipeline approach was developed to leverage the complementary strengths of both large language models and traditional machine learning techniques. In the first stage, GPT-4o-mini is used due to its speed, affordability, and good performance on focused tasks [15]. It analyzes game transcripts to identify the presence of 30 standardized ESRB content descriptors (mild violence, strong language, use of alcohol, etc) and one special descriptor (no_descriptors), which is exactly a focused task GPT-4o-mini prefers. GPT gives a binary feature vector (0 for absent, 1 for present descriptors) as a result. The approach makes use of GPT's contextual understanding capabilities to comprehend nuanced content presentations. In the second stage, a Random Forest classifier—trained on a dataset of over 2000 video games with their content descriptors and ratings—processes this binary feature vector (it is trained on vectors like that) to predict the appropriate ESRB rating (E, E10+, T, M).

The system architecture consists of a Python backend and React/TypeScript frontend. The backend handles transcript analysis, employing the GPT-4o-mini to process game text and identify content descriptors, followed by the trained Random Forest model for final rating prediction based on the descriptors GPT found. The system not only predicts the primary ESRB rating but also provides the second most probable rating with confidence scores. It also calculates category severity percentages for each content category (Violence, Language, Substances, Suggestive Content) and extracts evidence supporting the identified descriptors. Additionally, Gemini is used to generate a comprehensive game summary describing the plot, gameplay mechanics, and setting, providing context for the content analysis. The frontend presents this information in a visually appealing and interactive interface, allowing users to explore the analysis results in detail, including color-coded content severity visualizations, expandable evidence sections with transcript quotes, most representative words, and rating

confidence metrics.

The system includes several key functionalities. One of these is the ability to choose between Gemini and GPT for evidence extraction, allowing for more flexibility when analyzing game transcripts. While the context window size of GPT-4o-mini is 128,000 tokens, Gemini has a window of one million tokens, and it can handle much larger transcripts without having to cut them into segments, making it suitable for analyzing longer transcripts. A dedicated summarizer section was also implemented, allowing users to request specific aspects of the game to be summarized, such as gameplay, setting, plot, or game mechanics. This functionality can help with understanding the game better and not only see the concerning content. Moreover, a questionnaire section was added where users can directly input content descriptors, bypassing the LLM pipeline entirely and using only the Random Forest model for prediction. This alternative approach provides more reliable predictions in cases where the user already has knowledge about the game's content, making the system more versatile for different use cases.

Evaluation across three separate runs using 40 game transcripts demonstrated consistent accuracy of around 82% for the complete pipeline, with nearly perfect accuracy for E rating (93%) and perfect accuracy for M ratings (100%). Notably, the E10+ rating category presented the greatest challenge, with average accuracy rates of around 57% across the three runs. This challenge aligns with the inherent difficulty in distinguishing the subtle differences between content appropriate for children under 10 versus that for those 10 and older. The T rating had moderate consistency with around 77% accuracy. The system significantly outperforms both the BERT and Random Forest approaches (40% and 50% respectively) and standalone GPT predictions (70%).

Throughout development, several challenges were addressed. The most significant challenge was balancing GPT's tendency to be either too hesitant or too aggressive when flagging content descriptors. Left unaddressed, GPT would either flag virtually all concerning content descriptors (being overly aggressive) or miss important content indicators (being too hesitant). This was addressed by optimizing the GPT prompt with specific instructions. Another challenge was managing the computational requirements for analyzing lengthy game transcripts, which was resolved by implementing a segmented analysis approach that processes large transcripts in manageable chunks while maintaining contextual understanding across segments.

The next section will introduce the research related to this project.

■ Related Research (no more than 7 page)

As the gaming industry continues to grow, the automated prediction of video game age ratings has started to get attention. Due to the complexity of the problem, it is not the most popular topic of research, but there are still some studies. Two notable approaches in this domain have employed different methodologies to address the challenge of automating content evaluation: dictionary-based linguistic analysis and machine learning on pre-labeled content descriptors. This section will examine these studies, analyzing their methodologies, effectiveness, and limitations. After that, the analysis will be shifted to research related to LLM feature engineering for traditional ML, which is the main approach of this graduation work.

Maisto et al., in their research named "Extracting video games rating labels from transcript files," developed an unsupervised system to extract age-related content indicators from video game transcripts, focusing on PEGI and ESRB rating systems [13]. Their methodology combined specialized lexicons (e.g., dictionaries for slang, violence, drugs) with semantic expansion using the COALS algorithm, which uses distributional semantics to improve keyword lists with contextually related terms. They created several specialized dictionaries of varying sizes: a Dictionary of Slang (5,913 terms), Dictionary of Violence-related terms (1,173 terms), Dictionary of Drugs (546 terms), and Dictionary of Discriminatory terms (635 terms). These dictionaries were compiled from various online resources such as urbandictionary.com and theonlineslangdictionary.com, and included both simple words and multi-word expressions to improve semantic tagging accuracy.

The system calculated indices for Bad Language, Violence, and Drugs by normalizing dictionary matches against transcript length, supplemented by emotion intensity scores from the NRC Affect Intensity Lexicon. For example, the Bad Language Index in their study was calculated as the sum of Slang and Discrimination values, the Violence Index used Violence terms plus Anger, Fear, and Disgust emotion scores, while the Drugs Index was based on raw occurrence counts due to their relative rarity in the texts. For classification, transcripts were vectorized using TF-IDF, expanded semantically, and clustered via modularity-based network analysis with a resolution parameter of 0.9 to generate distinct game categories.

They analyzed a dataset of 50 video game transcripts from multiple release dates (1988-2020), genres, and target audiences, with a total of 701,384 tokens. The games they chose ranged from family-friendly titles like Super Mario

Party (rated E) to mature titles like *Cyberpunk 2077* (rated M). Their system was able to achieve 62% accuracy for exact rating matches and 79% for the presence/absence of content descriptors. It is worth noting that violence detection underperformed due to its visual-centric nature, while drug-related content showed higher accuracy (69%) due to its mention in the text. Their unsupervised classification approach using modularity algorithms produced eight distinct game classes like “Strong language action games” with 88% internal coherence. This showed that analyzing transcripts is possible without needing pre-labeled training data.

However, the transcript-based approach they implemented also has significant drawbacks. One major challenge is that a simple dictionary does not understand context. For example, their system treats every flagged word the same way, whether it is used in a humorous, critical, or narrative context. Their system also performed ineffectively with games containing limited dialogue, as prediction quality correlated directly with transcript volume. Very small transcripts (less than 1,000 tokens) like *SpongeBob SquarePants: SuperSponge* provided less accurate results with their approach. The researchers noted that this limitation would be particularly problematic for online games that often contain a minimal amount of text. The dictionaries they implemented required constant manual updating to understand evolving language patterns, especially slang and emerging terminology. Their system also struggled with fantasy or sci-fi games where fictional drugs with made-up names appeared, as these wouldn't match their real-world drug terminology database. Moreover, their research only used three concerning types of content (language, violence, drugs), ignoring critical ESRB descriptors like sexual themes, which limited the potential of their rating approach compared to official systems that make use of many more content categories.

On the other hand, Alomari et al., in their work named “Prediction of the Digital Game Rating Systems based on the ESRB,” took a different route by using machine learning on pre-labeled content descriptors [14]. The research involved a large dataset of 2,053 PlayStation 4 games, for which each game had 38 manually assigned content descriptors (such as “Violence” or “Strong Language”) marked simply as present or absent. By converting the content descriptors into binary features, they were able to train several classification models to predict the final game ratings. Their methodology strictly followed the CRISP-DM (Cross Industry Standard Process for Data Mining) approach, which provided a structured framework for the entire analysis process. The

dataset was carefully split into training (80%) and testing (20%) sets to ensure robust model evaluation, resulting in 1,642 games for training and 411 for validation.

Using RapidMiner software for implementation, they compared six different machine learning models: Generalized Linear Models (GLM), Decision Trees, Deep Learning, Gradient Boosted Trees, Random Forest, and Naive Bayes. Each model was evaluated using multiple performance metrics including accuracy, precision, recall, and F-score, both for overall performance and for each individual rating category (Everyone, Everyone 10+, Teen, and Mature). This comprehensive evaluation approach allowed them to understand each model's strengths and weaknesses across different rating classes.

Among the various models they tested, a Generalized Linear Model (GLM) performed the best, achieving an overall accuracy of 90.27%, outperforming Deep Learning (89.78%), Decision Tree (89.78%), Gradient Boosted (89.05%), Random Forest (83.7%), and Naive Bayes (77.86%). The GLM model showed particularly strong performance in classifying "Everyone" rated games, with a precision of 99% and an accuracy of 98.1%. It also performed well for "Mature" games with 96.1% accuracy, though the precision was somewhat lower at 82.7%.

One of the advantages of this method is that it clearly shows which content descriptors are most important in predicting ratings. Through feature importance analysis, they identified eight critical content descriptors with the highest predictive power: Violence (weight=1.0), Strong Language (0.84), Fantasy Violence (0.79), Blood (0.76), Sexual Themes (0.70), Blood and Gore (0.68), Intense Violence (0.66), and Mild Fantasy Violence (0.60). This finding confirmed that relatively few content descriptors drive most rating decisions in the ESRB system. Their analysis also produced a detailed weight chart for all 38 descriptors, showing that many descriptors like "Simulated Gambling" (0.017), "Tobacco Reference" (0.014), and "Sexual Violence" (0) had minimal impact on the final rating predictions.

The researchers provided detailed confusion matrices for each model, allowing for granular analysis of where classification errors occurred. For example, with the GLM model, 5 games that should have been rated as E10+ were mistakenly classified as E, while 8 games that should have been T-rated were incorrectly classified as E10+. These patterns helped identify the areas where rating distinctions are most challenging, particularly in the middle categories where content differences can be subtle.

Despite its high accuracy, this machine learning approach also has limitations. The reliance on pre-labeled content descriptors means that their system can only be applied to games that were already evaluated by human reviewers. This makes their approach not useful enough for new games or unreleased games that did not receive established content descriptors. Therefore, it can essentially only be used as a secondary classification layer rather than a primary content analysis system. Additionally, their system oversimplifies the content by not capturing the intensity or subtle differences in how a particular element is presented. The approach also does not consider the context in which these descriptors appear, treating each descriptor as a binary value without accounting for frequency, prominence, or narrative context.

Both studies highlight complementary strengths and fundamental challenges in automated game rating prediction. A promising solution would be to combine these two methods. A hybrid approach could use advanced natural language processing techniques, like those found in large language models, to automatically extract detailed content features from game transcripts. These features could then serve as inputs for a supervised machine learning model that predicts ratings with high accuracy.

The recent research in applying advanced language models alongside traditional machine learning techniques provides strong support for this work's two-stage pipeline approach. Three studies, spanning medical diagnostics, academic text analysis, and biomanufacturing prediction, offer clear evidence that combining GPT-based feature extraction with machine learning classifiers yields both high accuracy and transparency.

Heitz et al. demonstrated a novel approach to Alzheimer's disease detection by using GPT-4 to extract complex linguistic features from transcripts of spontaneous speech [16]. Instead of using GPT-4 directly for classification, it was used to identify and quantify five semantic features in speech that are clinically relevant to Alzheimer's disease but difficult to measure using traditional computational linguistics methods. These five critical features include word-finding difficulties, semantic paraphasias, syntactic simplification, impoverished vocabulary, and discourse impairment. Their evaluation was rigorous, involving comparisons with traditional linguistic features, validation against proxy measures, and agreement with human raters. The extracted features were then fed into a Random Forest classifier, which significantly improved detection accuracy, achieving an AUROC of 0.931 on manually transcribed data and maintaining robust performance on automatically generated

transcripts. This study shows that GPT can effectively extract complex, context-dependent features that traditional feature engineering methods often miss. The study supports the pipeline approach since it shows that by integrating features extracted by LLM with a robust classifier, we can achieve high performance even on complex, unstructured data.

Similarly, another research by Wang et al. proposed a system called "GPT-Signal" that applies a similar hybrid approach in the financial domain [17]. Their work focused on semi-automated feature engineering for financial signal generation, a task that usually relies on indicators made by people. Using GPT-4, the system analyzed existing financial signals and generated new, non-linear combinations that better predict stock returns. The authors designed a two-step prompt approach where GPT-4 first generated definitions and effects of existing signals, then created new signals based on these insights. Their methodology involved evaluating the generated signals through both correlation analysis and Fama-MacBeth regression to assess predictive power. Their experimental results across different sectors of S&P 500 companies demonstrated that the GPT-generated signals consistently outperformed baseline models that relied solely on traditional financial indicators. For example, their new "Efficiency Value Composite" signal showed a notably higher correlation with stock returns than any existing signals. This work provides strong validation that GPT-4 can effectively analyze complex financial data to generate meaningful features that, when incorporated into predictive models, yield superior performance to traditional feature engineering methods.

Moreover, Xiao et al. presented a workflow that is similar to this work's approach, using GPT-4 to extract knowledge from synthetic biology journal articles to support machine learning applications [18]. They developed a prompt engineering pipeline using GPT-4 to extract key information from journal articles focused on oleaginous yeasts. Their system processed 176 publications on *Yarrowia lipolytica* and *Rhodospiridium toruloides*, extracting detailed bioprocess conditions, metabolic pathways, and genetic engineering methods. It was then followed by human intervention to ensure data quality. The extracted data was organized into structured datasets with 2,037 instances, each containing both outputs (product titers) and inputs (feature variables). They were able to create structured datasets with feature selections that enabled machine learning approaches to predict fermentation titers. The Random Forest model they trained on these datasets achieved an R^2 of 0.86 on unseen test data, showing

the method's decency. The study also demonstrated the effectiveness of transfer learning, where knowledge from well-studied domains was applied to less-explored scenarios. This research provides further evidence that GPT can accelerate knowledge extraction from complex, technical literature and that the structured features derived from such an approach can effectively support traditional machine learning predictions.

Additionally, Yang et al., in their comprehensive scoping review titled "GPT for Games," examined 55 research articles to explore the potential applications of GPT models in game development and analysis [16]. Their review identified five key application areas: procedural content generation, mixed-initiative game design, mixed-initiative gameplay, playing games, and game user research. While none of the reviewed studies specifically addressed content rating prediction, their analysis revealed that GPT models excel at extracting context-dependent features from game text, particularly for narrative content analysis. The authors noted that smaller GPT models, when properly fine-tuned for specific gaming applications, can sometimes outperform larger general-purpose models in domain-specific tasks. This finding supports this work's approach of combining GPT's contextual understanding capabilities with traditional machine learning methods, as it suggests that specialized applications of language models to game content can be highly effective when properly implemented. Furthermore, their observation that game user research applications of GPT remain relatively unexplored highlights the novelty of using these models for automated content evaluation systems like the one developed in this graduation work.

Together, these studies demonstrate that combining GPT's feature extraction capabilities with traditional machine learning models like Random Forest represents a powerful approach across diverse domains. The GPT is able to capture complex information and transform it into structured features, while Random Forest and similar algorithms can provide transparent, accurate predictions based on these features.

■ Introduction of Proposal Work (no more than 7 page)

The original goal for this work was to create an interpretable and automated system that would be able to predict ESRB age rating for video games based on their transcript and provide some evidence. Due to BERT's bidirectional architecture, contextual embeddings, adaptability via fine-tuning, and effectiveness for tasks requiring deep semantic analysis [17], it was decided to use it as an initial approach for this work.

The implementation used RoBERTa-base, a robustly optimized BERT variant with 125 million parameters known for its improved training methodology and performance over the original BERT. First, 40 game transcripts were collected, representing games across all four main ESRB rating categories (E, E10+, T, M). These transcripts underwent extensive preprocessing through a custom pipeline that cleaned special characters, removed metadata labels, normalized whitespace, and ensured proper punctuation spacing. The cleaned transcripts were then divided into chunks of 128 tokens each using the BERT tokenizer to maintain semantic coherence while respecting model input limitations.

The training dataset contained 14,507 text chunks with a distribution of M: 4,413 samples (30.42%), E10+: 3,799 samples (26.19%), T: 3,690 samples (25.44%), and E: 2,605 samples (17.96%). This slight class imbalance was addressed during training by implementing class weights in the loss function to ensure the model would not be biased toward more frequently occurring classes. RoBERTa was fine-tuned using the Transformers library with a sequence classification head attached to the [CLS] token embedding. The hyperparameter optimization process explored learning rates between $1e-6$ and $1e-4$, batch sizes of 8, 16, or 32, epochs between 3 and 7, and weight decay values between 0.0 and 0.3. The final model used a learning rate of $1e-6$, batch size of 16, 5 epochs, and weight decay of 0.05 based on validation performance.

When evaluated on 20 game transcripts held out as a test set, the BERT approach achieved only 40% overall accuracy (8 out of 20 games correctly classified), with particularly poor performance on M-rated games (0% accuracy, 0/5 correct) while performing well on E10+ games (83.3% accuracy, 5/6 correct). E-rated games showed moderate performance (40% accuracy, 2/5 correct), while T-rated games were challenging (25% accuracy, 1/4 correct). The model showed clear signs of overfitting, with training loss decreasing substantially from 1.38 to 0.36 across epochs while validation loss actually increased from 1.36 to 1.72, indicating the model was memorizing training patterns rather than actually

learning generalizable features.

Detailed error analysis revealed that the model consistently misclassified M-rated games as T-rated, suggesting difficulty in distinguishing the severity levels of mature content. For example, in "yakuza0" (an M-rated game), only 28.6% of chunks were classified as M, while 49.2% were classified as T. This pattern repeated across other M-rated games, with "bloodrayne" showing 80.0% T predictions and only 12.0% M predictions. The model also struggled with chunk-level consistency, often making vastly different predictions for different sections of the same game, indicating an inability to maintain a coherent understanding of content across the entire transcript.

Additionally, a Random Forest approach using TF-IDF features was implemented as an alternative classification method. The implementation used a TF-IDF vectorizer with 5,000 maximum features, minimum document frequency of 2, and unigram and bigram features (ngram_range 1-2). The model combined predictions from both whole-game analysis and chunk-based predictions with a weighted average (0.7 for whole game, 0.3 for chunk predictions). This approach achieved approximately 50% accuracy on the same 20 game transcripts (10 out of 20 correctly classified). While marginally better than the BERT implementation, it still struggled with nuanced content classification, particularly with distinguishing between T and M ratings where contextual understanding of content severity is crucial.

Both approaches revealed fundamental limitations in directly applying text classification methods to the complex task of game rating prediction. The BERT model, despite its sophisticated language understanding capabilities, failed to grasp the nuanced differences between content severity levels and showed signs of memorization rather than generalization. The Random Forest approach, while slightly more successful, still couldn't effectively model the complex relationships between specific content elements and final ratings. These experiments demonstrated that direct text classification methods were insufficient, prompting the development of the two-stage pipeline approach that would combine the contextual understanding of large language models with the classification power of traditional machine learning techniques trained on content descriptors rather than raw text.

It was first decided to train a random forest classifier part of the pipeline. The dataset with video game ratings by ESRB was used to train this model [18]. The dataset contains 1895 games with 31 ESRB rating content descriptors with the name and console as features for each game. A single data

point represented as a binary value 0-1 for Console and a binary vector for the features of ESRB content. The dataset distribution included approximately 850 T-rated games and about 450 games for each of the other rating categories (E, E10+, and M). The 31 ESRB content descriptors served as the primary features for the Random Forest model. Each descriptor was represented as a binary value (0 for absent, 1 for present) in the game. This approach aligns with how ESRB assigns ratings in practice, where specific content elements determine the final rating category. The features such as name, console were removed, and the rating feature was used as a label (target variable). The dataset was split into training (80%) and validation (20%) sets.

Before training the model, standard preprocessing steps were applied. The data was first examined for missing values, though none were found in the binary descriptor columns. To ensure optimal model performance, feature scaling was implemented using `StandardScaler`, which normalized all features to have zero mean and unit variance. This preprocessing step was crucial for ensuring that all content descriptors contributed equally to the model's decision-making process, regardless of their frequency in the dataset. The scaled feature matrix was then used for both model training and evaluation.

Multiple classification algorithms were evaluated, including Logistic Regression, Decision Tree, and Random Forest. For each model, both default configurations and hyperparameter-tuned versions were tested across different random seeds to ensure reliability. Cross-validation with 10 folds was employed to provide robust performance estimates and prevent overfitting. The Random Forest classifier with default parameters consistently outperformed all other models, achieving an average accuracy of 85.20% across different random states with minimal standard deviation (0.0040), indicating exceptional stability. The tuned Random Forest achieved 84.87%, while Decision Tree and Logistic Regression models performed notably worse at 84.07% and 81.40%, respectively. On the final test set of 500 games, the Random Forest model achieved 85.4% accuracy. The classifier demonstrated strong performance across all ESRB categories, with particularly high precision for E ratings (95%) and M ratings (91%). While M-rated games showed high precision, they had a lower recall (67%), indicating that some M-rated games were misclassified into other categories.

The model's performance was analyzed in detail using various metrics. For E-rated games, the model achieved 95% precision and 95% recall (F1-score of 0.95). E10+ games showed 85% precision with 88% recall (F1-score of 0.86).

T-rated games had 79% precision and 88% recall (F1-score of 0.83), while M-rated games demonstrated 91% precision but only 67% recall (F1-score of 0.77). The overall weighted average F1-score was 0.85, which confirmed the model's strong performance across all rating categories.

Feature importance analysis was conducted to understand which content descriptors had the most significant impact on rating predictions. The analysis revealed that "strong_language" was the most influential feature with an importance score of 0.146, followed by "no_descriptors" (0.127), "fantasy_violence" (0.092), "blood" (0.088), and "blood_and_gore" (0.074). Other notable descriptors included "mild_fantasy_violence" (0.044), "suggestive_themes" (0.036), and "language" (0.034). This finding confirmed that violence and language-related descriptors play a crucial role in determining a game's age rating, aligning with ESRB's documented rating criteria.

While the Random Forest classifier demonstrated great accuracy in predicting ESRB ratings based on content descriptors, it requires these descriptors as input features to be already there for it. For new or unreleased games without pre-assigned descriptors, an automated method to extract these features from game transcripts was necessary. This challenge led to the development of a novel two-stage pipeline approach that leverages both large language models and traditional machine learning.

The first choice for selecting an LLM model was Gemini 2.0 Flash-Lite due to the preview version being free and a large context window size of one million [22]. However, no matter how the prompt was modified, Gemini was too aggressive with flagging content descriptors, which resulted in a low accuracy of about 40% for 40 transcripts when using a pipeline approach. The pipeline was predicting most of the E rated games as ET or T, which is why another approach had to be chosen.

After evaluating several approaches, GPT-4o-mini was selected as the first stage of the pipeline for automatically identifying content descriptors from game transcripts. GPT-4o-mini offers an optimal balance between accuracy and computational efficiency, making it suitable for production use. While larger models like GPT-4o or even GPT-4.5 might provide marginally better accuracy [23], they would significantly increase latency and API costs without proportional performance benefits for this focused task. Unlike dictionary-based approaches that simply match keywords, GPT models can understand nuanced contexts, distinguishing between mild fantasy violence and intense realistic violence, recognizing whether substances are merely referenced or actively consumed, and

determining if language is mildly inappropriate or explicitly vulgar. This context sensitivity is particularly important for game content analysis, where the same words or phrases might have different implications based on the surrounding narrative context. For example, GPT can distinguish between a character saying "I'll kill you" in a comedic cartoon setting versus a realistic combat scenario, which dictionary-based approaches would flag identically.

The implementation of the GPT-based descriptor detection involved careful prompt engineering. The system prompt was meticulously designed to guide the model toward accurate descriptor identification. It explicitly lists all 31 standardized ESRB content descriptors organized by category (Violence, Blood, Language, Sexual Content, etc.) to ensure the model searches for the complete range of possible descriptors. Importantly, the validation rules were specifically crafted to address a significant challenge: GPT models tend to be either overly hesitant (missing important content indicators) or too aggressive (exaggerating some content indicators). Multiple iterations of prompt engineering were required to achieve the optimal balance. In the Results section, the normal, hesitant, and this work's prompts will be provided as well as their performance.

The temperature of the GPT model was deliberately set to 0 for deterministic outputs, ensuring consistent and reproducible results across multiple evaluations of the same transcript. This design choice sacrifices creative variation for reliability, which is crucial for a rating system where consistency is paramount. The research by Rosol et al. suggests that a temperature set to 0 reduces randomness, and the model achieved higher consistency and accuracy with this temperature in structured exams [24]. This appears to be a common approach to increase reproducibility and reduce randomness, as in the study, Heitz et al. also used temperature 0 for this exact reason [16]. The system employs a structured JSON output format with a `content_flags` dictionary containing all 31 descriptors with binary values. This structured output facilitates seamless integration with the subsequent Random Forest classifier. By requiring a specific output format, the system ensures compatibility between the LLM analysis and the machine learning model without requiring complex parsing or transformation steps. This standardization also simplifies the API implementation, as the output can be directly used as input features for the classifier without additional processing.

Due to GPT having a small context window, for large game transcripts that exceed token limits, a segmented analysis approach was implemented. The system divides large transcripts into manageable segments of 125,000 tokens

each while maintaining context across them. This approach ensures that even massive game transcripts can be analyzed effectively while preserving contextual understanding throughout the document. For the largest game in the testing set, 3 segments were created, with each segment processed individually and the results combined.

The GPT analysis produces a standardized binary feature vector representing the presence or absence of each content descriptor, which serves as the input for the Random Forest classifier. This approach creates a clear separation of concerns: GPT handles the complex natural language understanding task of identifying content descriptors from raw text, while Random Forest, trained on a large dataset of games with known ratings, determines the appropriate age rating based on these descriptors.

Additionally, the system calculates "severity percentages" for each content category based on the severity of detected descriptors using a sophisticated algorithm that considers both the presence of descriptors and their relative intensity. For example, "intense_violence" contributes more to the Violence category score than "mild_violence," and having multiple descriptors within a category increases the overall severity rating. The algorithm employs a hierarchy system where only the most severe descriptor in each subcategory contributes to the final score, preventing inflation from multiple similar descriptors. These percentages provide an intuitive visual representation of content intensity across different dimensions.

To further enhance transparency and explainability, the system also extracts specific evidence from the transcript that supports the identified content descriptors. Here, the user is given the option to choose between GPT-4o-mini and Gemini for this evidence extraction process. Each has its advantages: GPT-4o-mini provides more concise and targeted evidence but is limited by its context window, while Gemini can process much larger portions of the transcript (up to 1 million tokens) without segmentation, providing more comprehensive evidence. The system identifies the three most representative words for each descriptor along with contextual examples from the transcript. This evidence extraction enables users to understand not just what content descriptors were detected, but exactly which parts of the transcript triggered them. The system also counts occurrences of each identified representative word throughout the transcript, providing frequency data.

The system also generates a comprehensive game summary that is around 100 words using Google's Gemini model, selected due to its larger

context window and efficient handling of lengthy text. This summary provides an overview of the game's plot, setting, gameplay mechanics, and overall atmosphere, giving additional context to the content analysis and helping users understand the broader narrative framework in which potentially concerning content appears.

In addition to the core analysis functionality, two complementary components were developed: a dedicated Summarizer and a Content Questionnaire. The Summarizer component allows users to request summaries that are around 250 words, with different focuses. Users can select from four focus areas: Gameplay, Setting & Atmosphere, Plot Points, and Game Mechanics. For each focus type, the system uses a specialized prompt designed to direct Gemini's attention toward the relevant aspects of the game transcript. This focused approach produces more detailed and useful summaries than general-purpose summarization would provide. The Summarizer's implementation ensures that even extremely large transcripts are handled efficiently by leveraging Gemini's extensive context window.

The Content Questionnaire component represents an alternative approach to content rating that bypasses the LLM pipeline entirely. This component is valuable when users already have knowledge about a game's content or when analyzing games with minimal text. The questionnaire presents yes/no questions organized into seven categories: Violence, Blood, Language, Sexual Content, Substances, Humor, and Gambling. Each question corresponds directly to one of the 30 ESRB content descriptors (there is no question for "no_descriptors," which is why it is not 31). The implementation includes progress tracking that monitors completion at both the category and overall levels. Once all questions are answered, the questionnaire generates a binary feature vector processed by the same Random Forest model used in the main pipeline, ensuring consistency in rating predictions regardless of the input method. This approach provides a reliable alternative when transcript analysis might be less effective or when users prefer a more direct input method. After completing the questionnaire, results appear similar to the Analyzer page, displaying the predicted ESRB rating, confidence percentage, and alternative rating. It shows the same color-coded content factor visualizations but omits transcript evidence and game summaries. This streamlined approach is ideal for quick assessments or early-stage game concepts.

■ Implementation and Analysis Result (no more than 7 page)

Table 1: Accuracy Results of ESRB Rating Prediction Across Three Evaluation Runs

Game Title	Year	ESRB Rating	GPT-only Accuracy	GPT+RF Accuracy
Everyone (E) Rating				
Blinx: The Time Sweeper	2002	E	100.0%	100.0%
DuckTales: Remastered	2013	E	100.0%	100.0%
Gravity Falls: Legend of the Gnome Gemulets	2015	E	100.0%	100.0%
Mega Man Legends	1997	E	100.0%	100.0%
Mischief Makers	1997	E	100.0%	100.0%
SpongeBob SquarePants Movie	2004	E	100.0%	100.0%
Super Mario RPG: Legend of the Seven Stars	1996	E	100.0%	66.7%
Super Paper Mario	2007	E	100.0%	66.7%
Super Princess Peach	2005	E	100.0%	100.0%
Up	2009	E	100.0%	100.0%
Everyone 10+ (ET) Rating				
Crash Bandicoot N. Sane Trilogy	2017	ET	0.0%	100.0%
Daxter	2006	ET	0.0%	100.0%
Lego DC Super-Villains	2018	ET	0.0%	100.0%
Monkey Island	1990	ET	0.0%	66.7%
Open Season	2006	ET	0.0%	0.0%
Over the Hedge	2006	ET	0.0%	33.3%
Portal 2	2011	ET	0.0%	100.0%
Shadow the Hedgehog	2005	ET	0.0%	0.0%
Spyro: The Eternal Night	2007	ET	0.0%	0.0%
The Legend of Zelda: Skyward Sword	2011	ET	0.0%	33.3%
Teen (T) Rating				
Mortal Kombat vs. DC Universe	2008	T	0.0%	100.0%
Final Fantasy VII Rebirth	2024	T	100.0%	100.0%
Final Fantasy XII	2006	T	100.0%	66.7%
Halo Infinite	2021	T	100.0%	100.0%
Horizon Forbidden West	2022	T	100.0%	100.0%
Marvel's Spider-Man	2018	T	100.0%	0.0%
Naruto Shippuden: Ultimate Ninja Storm 4	2016	T	100.0%	100.0%
Prince of Persia: The Sands of Time	2003	T	100.0%	100.0%
Street Fighter V	2016	T	0.0%	0.0%
Warcraft III: The Frozen Throne	2003	T	100.0%	100.0%
Mature 17+ (M) Rating				
Assassin's Creed II	2009	M	100.0%	100.0%
BloodRayne	2002	M	100.0%	100.0%
F.E.A.R.	2005	M	100.0%	100.0%
God of War	2018	M	100.0%	100.0%
Grand Theft Auto: San Andreas	2004	M	100.0%	100.0%
Hotline Miami	2012	M	100.0%	100.0%
Metal Gear Solid V: The Phantom Pain	2015	M	100.0%	100.0%
Mortal Kombat X	2015	M	100.0%	100.0%
Resident Evil Village	2021	M	100.0%	100.0%
Yakuza 0	2015	M	100.0%	100.0%

The system was evaluated through three separate runs using a diverse set of 40 video game transcripts spanning from 1990 to 2024, ensuring a comprehensive and reliable assessment across different gaming eras, genres, and intended audience demographics. As shown in Table 1, the evaluation included

10 games per each rating. This table shows the average accuracies across three runs only using GPT and using the GPT+RF pipeline per each game.

Table 2: Summary of Accuracy by Rating Category Across Three Evaluation Runs

Rating Category	GPT-only Accuracy	GPT+RF Accuracy
Everyone (E)	100.0%	93.3%
Everyone 10+ (ET)	0.0%	56.7%
Teen (T)	80.0%	76.7%
Mature (M)	100.0%	100.0%
Overall	70.0%	81.7%

The two-stage pipeline combining GPT-4o-mini and Random Forest achieved 81.7% overall accuracy across three runs, significantly outperforming both the GPT-only approach (70.0%) and using only Random Forest with TF-IDF features (approximately 50%). This validates the hybrid approach that leverages both GPT's contextual understanding and Random Forest's structured classification..

It is important to acknowledge a potential methodological consideration: since many of the test games were released before GPT-4o-mini's knowledge cutoff (October 2023), one might assume the model could simply recognize the games and recall their official ESRB ratings rather than performing actual content analysis. However, the system demonstrated strong performance even on games released after this cutoff date. For example, "Final Fantasy VII Rebirth," released in February 2024 (well after GPT's knowledge cutoff), was correctly classified with 100% accuracy across all three runs. This suggests that the system is genuinely analyzing content rather than merely retrieving memorized ratings, validating the robustness of the approach for new and future game releases.

The system achieved perfect accuracy (100%) for M-rated games across all runs. This performance comes from the distinct nature of M-rated content, which typically contains unambiguous examples of intense violence, strong language, or mature themes that are easily detectable in transcripts. Similarly, the system performed excellently on E-rated games (93.3% accuracy). The few errors occurred with games like "Super Mario RPG" and "Super Paper Mario," where fantasy elements were occasionally misinterpreted as mild fantasy violence, leading to stricter rating predictions and them being predicted as "ET".

The Teen rating category showed solid performance (76.7%), though with some misclassifications. Games like "Marvel's Spider-Man" and "Street Fighter V" proved challenging, indicating difficulty in calibrating the severity of content that falls between mild and intense. This is because the system relies solely on textual information. Based on the text, "Marvel's Spider-Man" violence is mild,

which leads it to being predicted as "ET", but graphically it is quite moderate. A similar thing is happening with "Street Fighter V", text-wise it is intensely violent, which leads it to being predicted as "M", but graphically it is very moderate and somewhat cartoony.

The most challenging category was Everyone 10+ (ET), achieving only 56.7% accuracy. This aligns with expectations, as distinguishing between E and ET involves subtle considerations difficult to capture through text analysis. Still, GPT alone completely failed to identify ET-rated games (0% accuracy).

In the first and the third run, the total average accuracy was 82.5%, while the second run was 80%. The consistency across three runs demonstrates the system's reliability, with minimal variance between runs indicating stable and reproducible results. The system showed consistent accuracy regardless of a game's release year, demonstrating robustness to evolving gaming content and language patterns over the 34 years.

To further evaluate the system's performance under different conditions, various temperature settings for the GPT model were tested to determine the optimal configuration. The temperature parameter in large language models controls the randomness in their outputs, with lower values producing more deterministic responses and higher values introducing more variability. Table 3 presents a comprehensive comparison of accuracy across different temperature settings for each ESRB rating category.

Table 3: Accuracy Comparison Across Different Temperature Settings

Temperature	E	ET	T	M	Overall Accuracy
0.0	90.00% (9/10)	60.00% (6/10)	80.00% (8/10)	100.00% (10/10)	82.50%
0.25	100.00% (10/10)	40.00% (4/10)	70.00% (7/10)	100.00% (10/10)	77.50%
0.5	90.00% (9/10)	30.00% (3/10)	80.00% (8/10)	90.00% (9/10)	72.50%
0.75	70.00% (7/10)	50.00% (5/10)	50.00% (5/10)	90.00% (9/10)	65.00%
1.0	50.00% (5/10)	30.00% (3/10)	30.00% (3/10)	60.00% (6/10)	42.50%

As shown in Table 3, temperature setting had a significant impact on the system's performance. A temperature of 0.0 achieved the highest overall accuracy at 82.50%, with particularly strong performance for M-rated games (100%), T-rated games (80%), and E-rated games (90%). The performance gradually declined as temperature increased, with overall accuracy dropping to

77.50% at 0.25, 72.50% at 0.5, 65.00% at 0.75, and only 42.50% at 1.0. This clear trend demonstrates that increasing randomness in the GPT model's outputs negatively impacts rating prediction accuracy.

A temperature of 0.0 provides not only the best performance but also ensures consistent and reproducible results across multiple runs, which is crucial for a reliable rating system. The deterministic nature of this setting eliminates variability in content descriptor identification, creating a more stable foundation for the Random Forest classifier in the second stage of the pipeline.

To analyze the impact of prompt engineering on system performance, three distinct prompt approaches were tested: an original balanced prompt, a minimal prompt with reduced instructions, and a hesitant prompt designed to minimize false positives. Table 4 presents a comprehensive comparison of accuracy across these different prompt engineering approaches for each ESRB rating category.

Table 4: Accuracy Comparison Across Different Prompt Engineering Approaches

Prompt	E	ET	T	M	Overall Accuracy
Original (Balanced)	90.00% (9/10)	50.00% (5/10)	80.00% (8/10)	100.00% (10/10)	80.00%
Minimal	10.00% (1/10)	40.00% (4/10)	70.00% (7/10)	40.00% (4/10)	40.00%
Hesitant	100.00% (10/10)	10.00% (1/10)	10.00% (1/10)	20.00% (2/10)	35.00%

As shown in Table 4, the prompt engineering approach significantly influenced the system's accuracy. The original balanced prompt achieved the highest overall accuracy at 80.00%, with particularly strong performance for M-rated games (100%), E-rated games (90.00%), and T-rated games (80.00%). In contrast, both alternative approaches performed substantially worse, with the minimal prompt achieving only 40.00% overall accuracy and the hesitant prompt performing even more poorly at 35.00% overall.

The hesitant prompt demonstrated an interesting pattern where it achieved perfect accuracy for E-rated games (100%) but performed extremely poorly across all other categories, with only 10.00% accuracy for ET and T ratings, and 20.00% for M ratings. This indicates that the overly hesitant approach effectively biased the system toward the least restrictive rating category. This is because it prefers not to flag most of the content descriptors that are not that common.

Meanwhile, the minimal prompt showed more balanced but generally

poor performance across categories, with its highest accuracy for T-rated games (70.00%). The minimal prompt is generally more aggressive, so it finds the content descriptors that are not really present there, which dramatically reduces the accuracy for E-rated games as it can be seen on the table. These findings demonstrate that comprehensive, well-balanced instructions in the system prompt are crucial for accurate content analysis. The original prompt's detailed validation rules and thorough content descriptor explanations provide the necessary guidance for the GPT model to correctly identify and classify game content without being too aggressive or too hesitant in flagging potentially concerning material.

To make the system accessible and user-friendly, a comprehensive web application was developed with an intuitive interface. The application features a modern, responsive design with switable dark/light modes (shown in the figures) to accommodate user preferences.

The interface is organized into three main sections accessible via the navigation bar: Analyzer (the primary tool), Summarizer, and Questionnaire. In the Analyzer section, users can input game transcripts in a responsive text area that automatically expands as content is added and displays the current word count in real-time. Users can select between GPT and Gemini models for evidence collection before submitting their transcript for analysis.

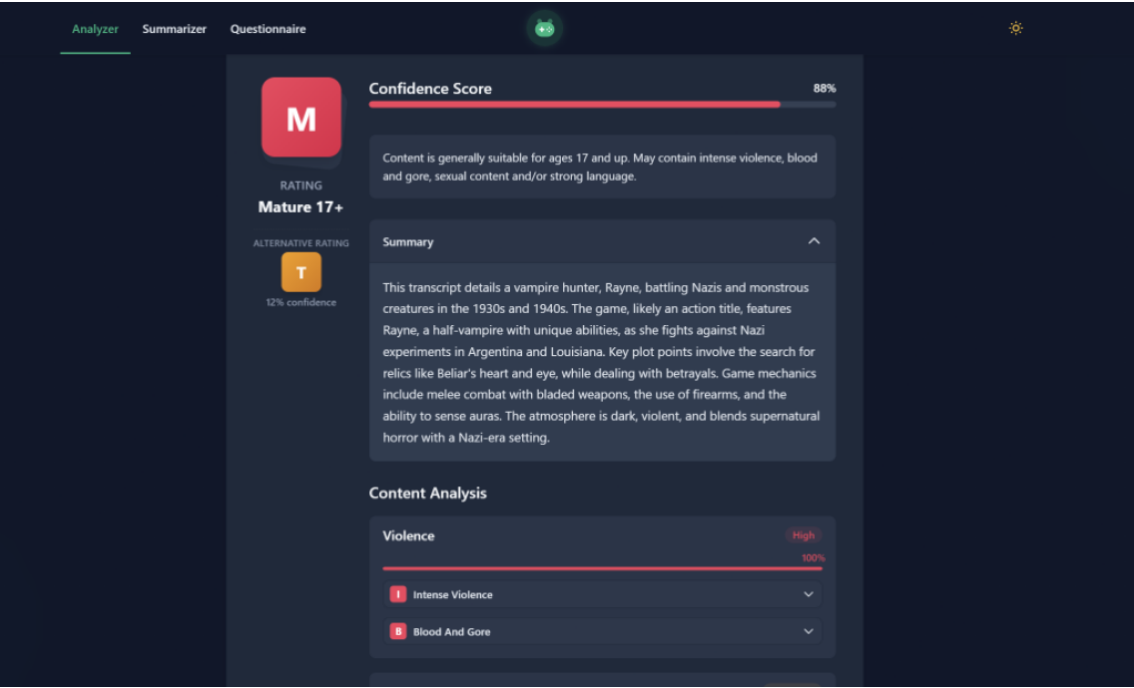


Figure 1: Main Analysis Interface Showing Rating and Content Category Overview

The results section prominently displays the predicted ESRB rating (in

Figure 1, "M" for Mature 17+) with an accompanying confidence score (88%) represented by a progress bar. The system also provides an alternative rating prediction (T/Teen in this example) with its confidence percentage (12%), acknowledging the inherent uncertainty in some classification tasks. A formal description of the rating appears below, explaining its typical content characteristics.

An important feature of the interface is the collapsible Summary section, which presents a concise overview of the game generated by Google's Gemini model. The Content Analysis section visualizes the severity percentages for each content category (Violence shown at 100% in Figure 1) using color-coded progress bars that intuitively communicate content intensity. For each category, the system displays specific content descriptors identified in the transcript, such as "Intense Violence" and "Blood and Gore" shown in Figure 1.



Figure 2: Expanded Content Descriptor View with Evidence and Word Analysis

Figure 2 demonstrates the system's transparency features when a user expands a content descriptor. Each descriptor includes a formal definition and, critically, evidence extracted directly from the transcript that justifies its identification. For instance, the "Intense Violence" descriptor is supported by quotes containing graphic violent content, while "Blood and Gore" is evidenced

by explicit descriptions of bodily harm.

The evidence section also provides "Top Words" related to each descriptor, displaying the terms most strongly associated with that content type along with their frequency in the transcript. While some words show a count of 0 as seen in Figure 2 (e.g., "stab (0)"), this occurs because GPT identifies base forms of words while the counting mechanism searches for exact matches. For example, "stab" is identified in the phrase "Bloodrayne stabs her in the neck," but the exact form "stab" doesn't appear.

In addition to the main Analyzer, the system includes two complementary components: Summarizer and Questionnaire.

The Summarizer page allows users to generate focused game summaries from transcripts. Users can select from four focus areas: Gameplay, Setting & Atmosphere, Plot Points, and Game Mechanics. Each option directs Gemini's attention toward specific aspects of the game content. The interface displays the transcript at the top, followed by focus selection buttons arranged in a 2×2 grid. After processing, a concise summary (approximately 250 words) appears below, with a header indicating the chosen focus. The Summarizer leverages Gemini's extensive context window to handle even the largest game transcripts without segmentation.

The Questionnaire offers an alternative approach to content rating that bypasses the LLM pipeline entirely. The interface presents yes/no questions organized into seven categories: Violence, Blood, Language, Sexual Content, Substances, Humor, and Gambling. Each question corresponds directly to one of the 30 ESRB content descriptors, with detailed explanations. A progress bar tracks completion at both category and overall levels. Upon completion, the system displays the predicted ESRB rating with confidence percentage, alternative rating, and color-coded content factor visualizations - ideal for quick assessments or games with minimal text.

The expandable interface elements throughout the application allow users to focus on specific aspects of the analysis while maintaining a clean, uncluttered view of the overall results. This careful balance between comprehensive information and usability ensures that the application remains accessible to various users, from game developers seeking early rating estimates to parents researching game content appropriateness for their children. The system provides value across the gaming ecosystem by making content evaluation more transparent and evidence-based.

■ Conclusion and Impression (no more than 2 page)

This graduation work successfully developed an automated system for predicting ESRB ratings of video games based solely on their transcripts. The novel two-stage pipeline approach, combining the contextual understanding capabilities of GPT-4o-mini with the classification power of a Random Forest model, achieved 81.7% overall accuracy across three evaluation runs using 40 diverse game transcripts spanning from 1990 to 2024. This hybrid methodology significantly outperformed both the standalone GPT approach (70.0%) and the initial BERT implementation (40%), validating the effectiveness of leveraging complementary strengths from different AI paradigms. Most notably, the system demonstrated perfect accuracy for Mature-rated games (100%) and excellent performance for Everyone-rated games (93.3%), confirming its effectiveness at distinguishing content suitable for the youngest audiences from distinctly adult content.

The extensive experimentation with temperature settings and prompt engineering approaches demonstrated the critical importance of finding the right balance in content flagging. The system achieved optimal performance with a temperature of 0.0, which provided not only the highest accuracy (82.5%) but also ensured consistent and reproducible results. The original balanced prompt significantly outperformed both minimal and hesitant alternatives, highlighting how crucial comprehensive, well-calibrated instructions are for accurate content analysis. Too cautious an approach severely undermined accuracy by failing to identify concerning content, while minimal guidance led to overly aggressive content flagging. This underscores the nuanced nature of content evaluation and validates the careful design choices made in the system's implementation.

Despite its strengths, the system faces challenges with the Everyone 10+ category, achieving only 56.7% accuracy. This limitation reflects the inherent difficulty in distinguishing subtle differences between content appropriate for children under versus over 10 years old – a challenge that even human reviewers often struggle with. This category represents the smallest step between rating levels in the ESRB system, making it particularly challenging to classify through automated means. Still, since GPT alone completely fails to identify ET-rated games (0% accuracy) it demonstrates the value of the two-stage pipeline approach.

The Web application development presented its own set of challenges and learning opportunities. Creating an interface that balanced comprehensive information display with usability concerns required careful consideration of user

experience principles. The expandable evidence sections, color-coded content severity visualizations, and alternative rating predictions with confidence metrics all contribute to a transparent system that users can trust and understand. The addition of complementary components like the Summarizer and Questionnaire further enhanced the system's utility across different use cases, from detailed content analysis to quick preliminary assessments.

This graduation work has been a profound learning experience in combining advanced AI technologies with traditional machine learning approaches to solve a complex real-world problem. The process demanded meticulous attention to prompt engineering, extensive experimentation with various approaches, and creative problem-solving to overcome the limitations of individual technologies. Moving from the initial BERT implementation through multiple iterations to the final two-stage pipeline required persistence and adaptability, especially when promising approaches failed to deliver adequate results. I had to understand and navigate the strengths and weaknesses of different models – BERT's contextual understanding but tendency to overfit on limited data, GPT's powerful comprehension but vulnerability to biased responses without careful prompting, and Random Forest's classification strength when provided with the right features.

The financial investment of \$35 USD for GPT API testing represented a commitment to thorough validation and optimization. This relatively modest expenditure enabled comprehensive testing across different prompt variations, temperature settings, and game transcripts, ultimately contributing to the system's impressive performance. The investment was essential for tuning the system to achieve the right balance between overly hesitant and excessively aggressive content flagging.

Looking ahead, the automated content rating system developed in this graduation work demonstrates significant potential for practical application in the gaming industry. As the volume of released games continues to grow exponentially and content becomes increasingly complex, efficient and consistent evaluation tools like this will become increasingly valuable. While the system is not intended to replace human reviewers entirely, it offers a powerful complementary tool that could make age-appropriate content evaluation more accessible, consistent, and evidence-based. The methodology developed here could also be extended to other content rating systems beyond ESRB or adapted for additional media formats beyond video games, suggesting promising directions for future research and development.

■ References (no more than 1 page)

- [1] J. Batchelor, "GamesIndustry.biz presents... The Year In Numbers 2024," GamesIndustry.biz. Accessed: Mar. 26, 2025. [Online]. Available: <https://www.gamesindustry.biz/gamesindustrybiz-presents-the-year-in-numbers-2024>
- [2] J. Clement, "Global video game users 2029," Statista. Accessed: Mar. 26, 2025. [Online]. Available: <https://www.statista.com/statistics/748044/number-video-gamers-world/>
- [3] "Steam Game Release Summary by Year," SteamDB. Accessed: Mar. 26, 2025. [Online]. Available: <https://steamdb.info/stats/releases/?tagid=0>
- [4] "E for Everyone Was Assigned Most Often in 2024," ESRB. Accessed: Mar. 26, 2025. [Online]. Available: <https://www.esrb.org/blog/e-for-everyone-was-assigned-most-often-in-2024/>
- [5] "2021 Essential Facts About the Video Game Industry," Entertainment Software Association. Accessed: Mar. 26, 2025. [Online]. Available: <https://www.theesa.com/2021-essential-facts-about-the-video-game-industry/>
- [6] "Our History," ESRB. Accessed: Mar. 26, 2025. [Online]. Available: <https://www.esrb.org/history/>
- [7] "ESRB Ratings," ESRB. Accessed: Mar. 26, 2025. [Online]. Available: <https://www.esrb.org/>
- [8] "ESRB ratings process for physical and digital video games," ESRB. Accessed: Mar. 26, 2025. [Online]. Available: <https://www.esrb.org/ratings/ratings-process/>
- [9] J. Brightman, "Indies feel the pinch from Sony's new ESRB mandate," GamesIndustry.biz. Accessed: Mar. 26, 2025. [Online]. Available: <https://www.gamesindustry.biz/indies-feel-the-pinch-from-sonys-new-esrb-mandate>
- [10] "Content Moderation in a New Era for AI and Automation," Oversight Board. Accessed: Mar. 26, 2025. [Online]. Available: <https://www.oversightboard.com/news/content-moderation-in-a-new-era-for-ai-and-automation/>
- [11] J. R. Jim, M. A. R. Talukder, P. Malakar, M. M. Kabir, K. Nur, and M. F. Mridha, "Recent advancements and challenges of NLP-based sentiment analysis: A state-of-the-art review," *Natural Language Processing Journal*, vol. 6, p. 100059, Mar. 2024, doi: 10.1016/J.NLP.2024.100059.
- [12] "Will AI Models Ever Understand Context? The New Frontier of Deep Learning in Contextual Awareness," Hyperight. Accessed: Mar. 26, 2025. [Online]. Available: <https://hyperight.com/will-ai-models-ever-understand-context-new-frontier-of-deep-learning-in-contextual-awareness/>
- [13] A. Maisto, G. Martorelli, A. Paone, and S. Pelosi, "Extracting video games rating labels from transcript files," *Internet of Things*, vol. 16, Dec. 2021, doi: 10.1016/J.IOT.2021.100439.
- [14] K. Alomari, A. Q. Alhamad, H. Mobaideen, and S. Salloum, "Prediction of the Digital Game Rating Systems based on the ESRB," *Opcion*, vol. 35, pp. 1368-1393, Mar. 2019.
- [15] "GPT-4o mini," OpenAI. Accessed: Mar. 27, 2025. [Online]. Available: <https://platform.openai.com/docs/models/gpt-4o-mini>
- [16] J. Heitz, G. Schneider, and N. Langer, "Linguistic Features Extracted by GPT-4 Improve Alzheimer's Disease Detection based on Spontaneous Speech," in *Proceedings of the 31st International Conference on Computational Linguistics*, O. Rambow, L. Wanner, M. Apidianaki, H. Al-Khalifa, B. Di Eugenio, and S. Schockaert, Eds., Abu Dhabi, UAE: Association for Computational Linguistics, Jan. 2025, pp. 1850-1864. [Online]. Available: <https://aclanthology.org/2025.coling-main.126/>
- [17] Y. Wang, J. Zhao, and Y. Lawryshyn, "GPT-Signal: Generative AI for Semi-automated Feature Engineering in the Alpha Research Process," in *Proceedings of the Eighth Financial Technology and Natural Language Processing and the 1st Agent AI for Scenario Planning*, C.-C. Chen, T. Ishigaki, H. Takamura, A. Murai, S. Nishino, H.-H. Huang, and H.-H. Chen, Eds., Jeju, South Korea: -, Mar. 2024, pp. 42-53. [Online]. Available: <https://aclanthology.org/2024.finnlp-2.4/>
- [18] Z. Xiao, W. Li, H. Moon, G. W. Roell, Y. Chen, and Y. J. Tang, "Generative Artificial Intelligence GPT-4 Accelerates Knowledge Mining and Machine Learning for Synthetic Biology," *ACS Synth Biol*, vol. 12, no. 10, pp. 2973-2982, 2023, doi: 10.1021/acssynbio.3c00310.
- [19] D. Yang, E. Kleinman, and C. Harteveld, "GPT for Games: A Scoping Review (2020-2023)," in *2024 IEEE Conference on Games (CoG)*, IEEE, Aug. 2024, pp. 1-8. doi: 10.1109/cog60054.2024.10645548.
- [20] M. Koroteev, "BERT: A Review of Applications in Natural Language Processing and Understanding," Mar. 2021. doi: 10.48550/arXiv.2103.11943.
- [21] "Video Games Rating By 'ESRB,'" Kaggle. Accessed: Mar. 28, 2025. [Online]. Available: <https://www.kaggle.com/datasets/imohtn/video-games-rating-by-esrb/data>
- [22] "Gemini 2.0 Flash-Lite," Google DeepMind. Accessed: Mar. 28, 2025. [Online]. Available: <https://deepmind.google/technologies/gemini/flash-lite/>
- [23] "Models," OpenAI. Accessed: Mar. 28, 2025. [Online]. Available: <https://platform.openai.com/docs/models>
- [24] M. Rosoř, J. S. Gąsior, J. Łaba, K. Korzeniewski, and M. Młyńczak, "Evaluation of the performance of GPT-3.5 and GPT-4 on the Polish Medical Final Examination," *Sci Rep*, vol. 13, no. 1, p. 20512, 2023, doi: 10.1038/s41598-023-46995-z.

The code link: <https://github.com/freedom764/Graduation-Project>